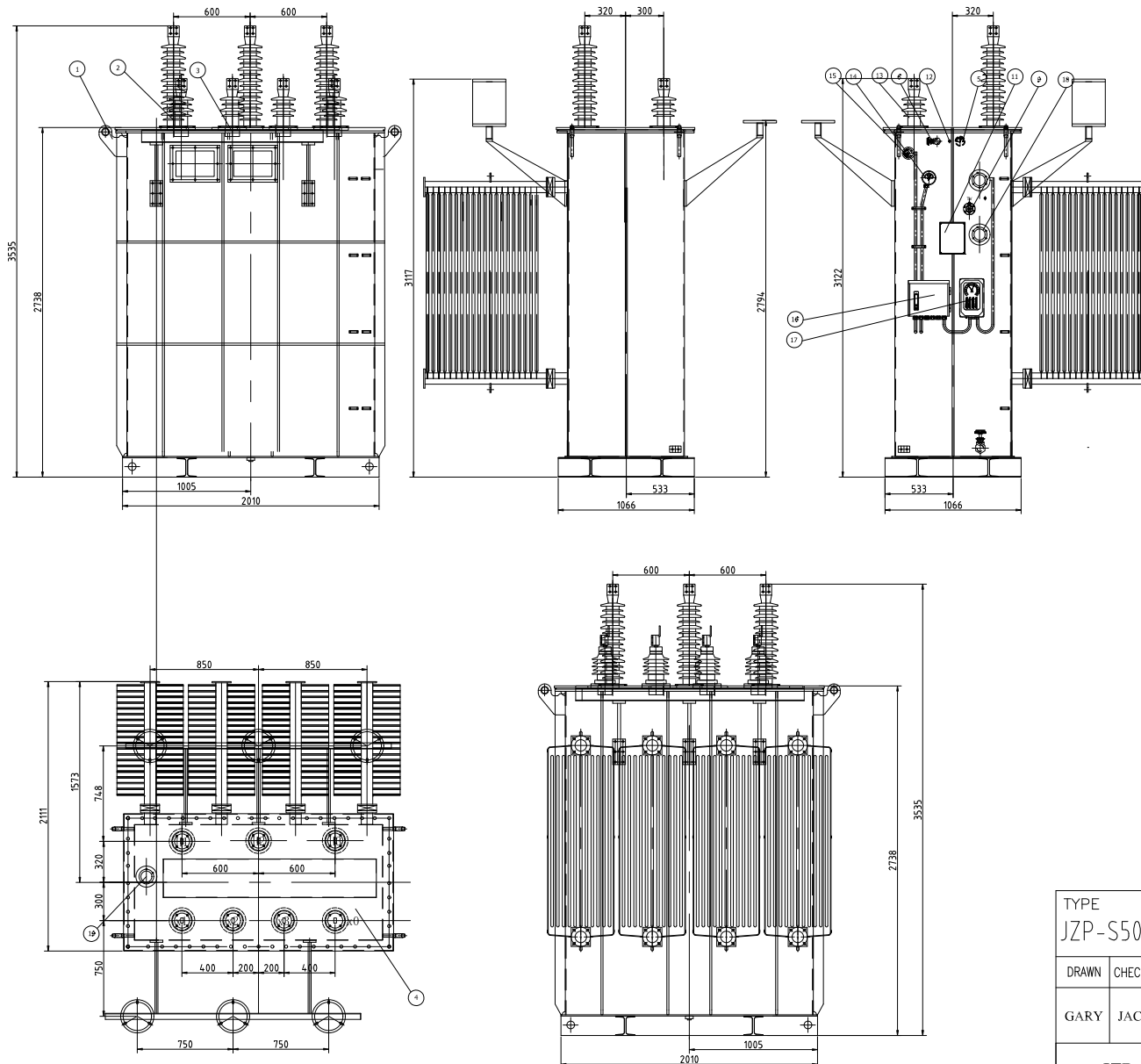


ON
DWG JZP-S5000-240306-001



TECHNICAL DATA

RATING	5000kVA
PRIMARY VOLTAGE	44000V
SECONDARY VOLTAGE	4160Y/2400V
FREQUENCY	60Hz
VECTOR GROUP	Dyn11
WINDING MATERIAL	COPPER
LOAD LOSS AT RATED VOLTAGE	33000W
NO LOAD LOSSES	5500W
IMPEDANCE	5.37%
ALTITUDE	≤1000m
No. OF TAPPOSITION	5
COLOR	ANSI GREY#70

ITEM	WEIGHT
TANK & ACCESORIES	8350 LBS
TOTAL OIL WEIGHT	2800L
TOTAL WEIGHT	26880LBS

PARTS LIST

ITEM	DESCRIPTION	QTY
1	LIFTING LUG	4
2	HV BUSHING 46KV 417A	3
3	COVER	1
4	HAND HOLE	2
5	VACUUM PRESSURE GAUGE	1
6	FILL PLUG	1
7	DRAIN VALVE WITH SAMPLING FLUG	1
8	RADIATORS PC-1000/12-520	4
9	TAP CHANGER(WSLI3 125/46-6×5)	1
10	GROUND TERMINAL	3
11	NAMEPLATE	1
12	PRESSURE RELIEF VALVE	1
13	LV BUSHING 15KV 1000A	4
14	OIL TEMPERATURE GAUGE w/2 FORM C CONTACT	1
15	OIL LEVEL GAUGE w/1 FORM C CONTACT	1
16	NEMA 4R BOX	1
17	WINDING TEMPERATURE INDICATOR	1
18	RAPID PRESSURE RISE RELAY WITH ALARM AND TRIP CONTACTS	1
19	EMERGENCY PRESSURE RELIEF DEVIC (QUALITROL) 208-60U	1

TYPE
JZP-S5000/44/4.16

DWG. NAME
General Assembly Drawing

DRAWN	CHECKED	CHECKED	APPROVED
GARY	JACK	IRIS	SAM

FIRST ANGLE
PROJECTION

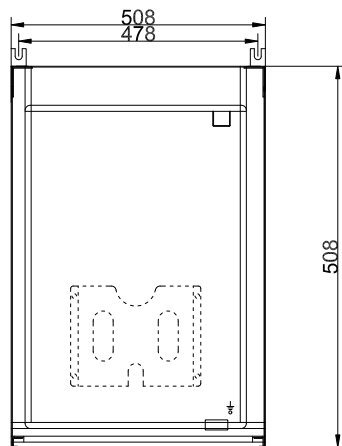
SCALE 1

JZP-S5000-240306-001

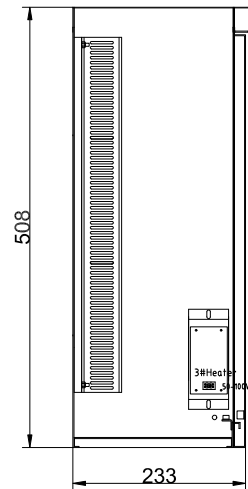
JZP
POWER TRANSFORMER

REV.
NO.

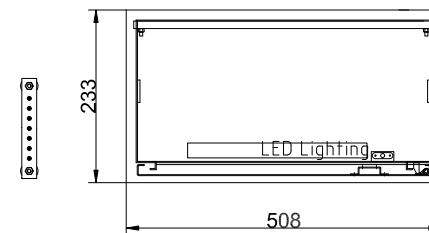
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Front View



Side View



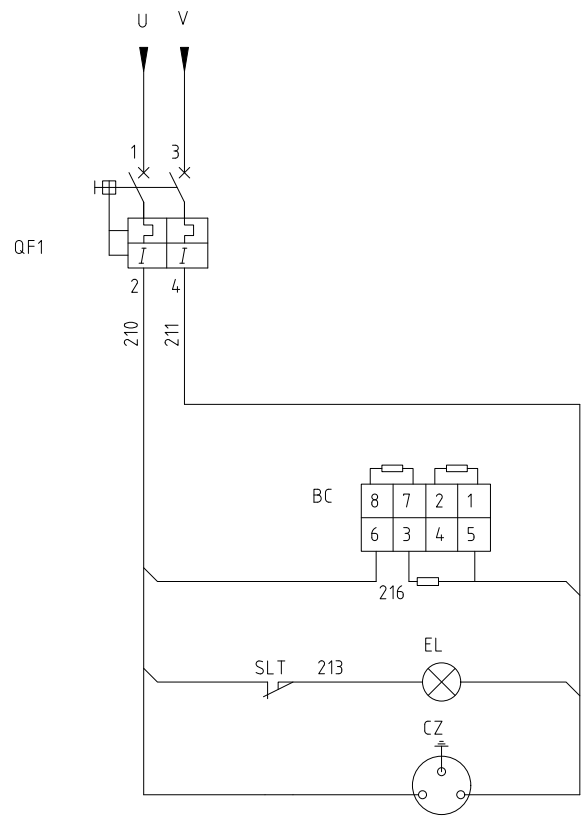
Top View

Technical conditions

1. The control box is wall mounted.
2. The material of the control box is 2.0mm stainless steel plate.
3. The protection grade of this control box is NEMA 4X .

Outline drawing of control box	BDKF-MX2F-01		
	Mark	Weight	Ratio
	A3		
	Total		page

Heating and lighting circuit



Electrical principle	BDKF-MX2F-03			
	Mark	Weight	Ratio	
	A3			
Total		Page		

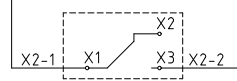


X1 LR-4.16
CT

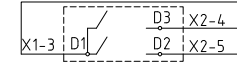
IS1 X1-1
IS2 X1-2

X1:CT				
Contact	Loop No.	No.	Contact function	Loop No.
IS1	1			27
IS2	2			28
	3			
	4			
	5			

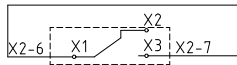
Pressure relief valve



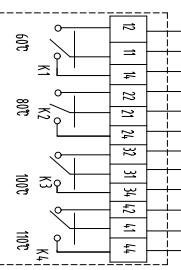
Oil level gauge



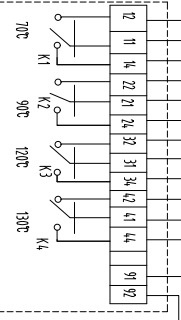
Pressure/vacuum gauge



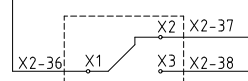
Oil level thermostat:BWY-804



Winding thermometer:BW-4

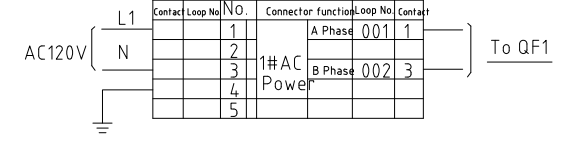


Pressure burst relay



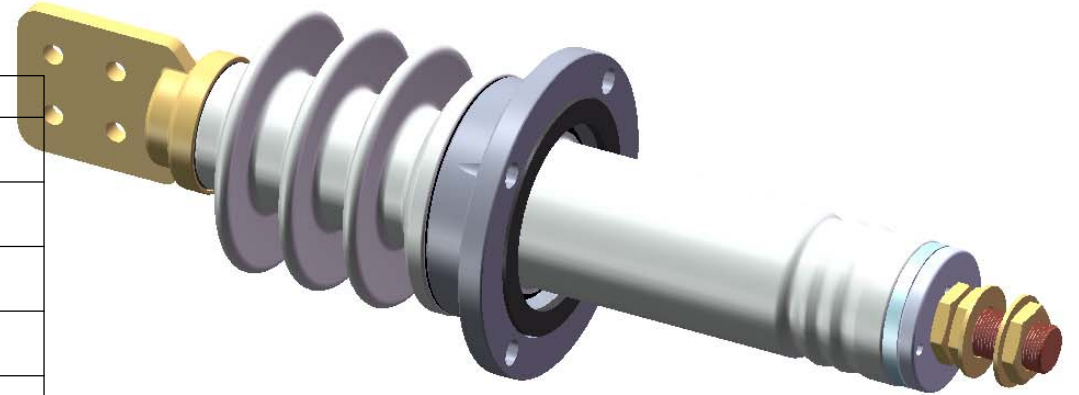
X2:				
Contact	Loop No.	No.	Contact function	Loop No.
X1	1			
X3	2			
D1	3			
D3	4			
D2	5			
X1	6			
X3	7			
	8			
	9			
	10			
	11			
	12			
	13			
	14			
	15			
	16			
	17			
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	35			
	36			
	37			
	38			
	39			
	40			
	41			
	42			
	43			
	44			
	45			

X3:AC power terminal				
Contact	Loop No.	No.	Contact function	Loop No.
	1		A Phase	001
	2		1#AC Power	
	3		B Phase	002
	4			
	5			



To QF1

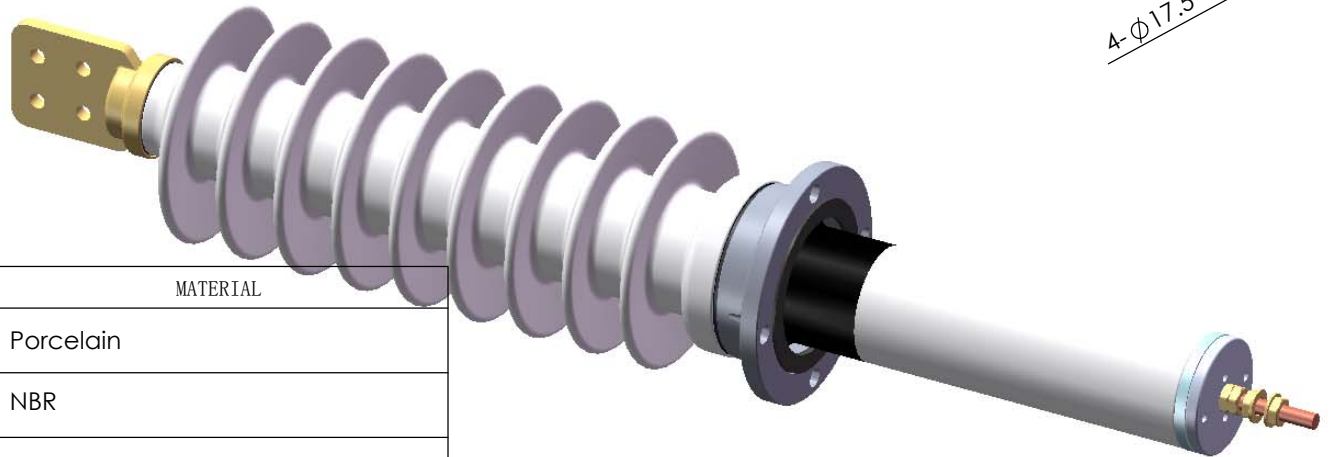
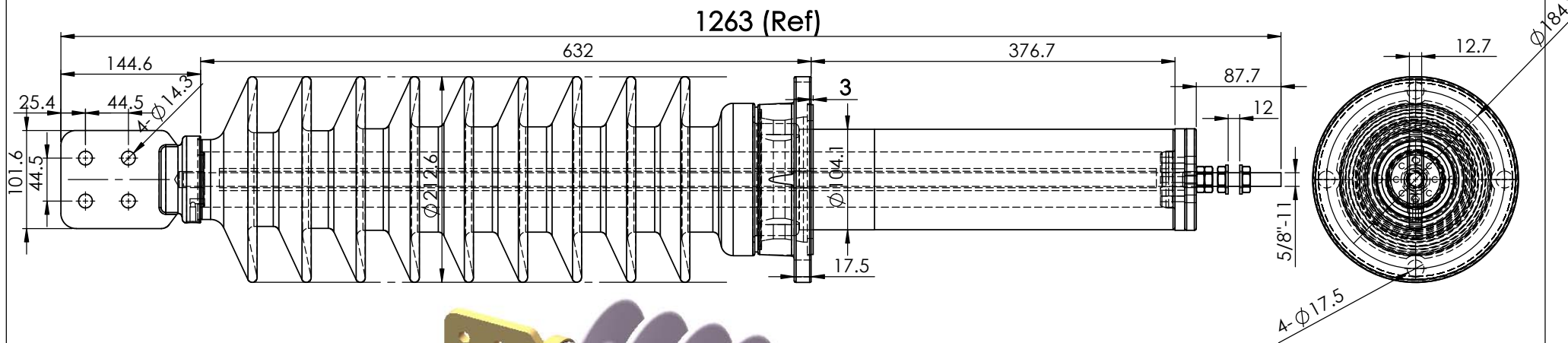
Output Signal	BDKF-MX2F-06			
	Mark	Weight	Ratio	
	A3			



Exposed copper and brass parts tinned, thickness >5 um

Rated Voltage	=34.5 kV ✓
Rated Current	✓ =1227 A
Creep Distance	>=366 mm
Impulse Level	=110 kV ✓
dry withstand Voltage	=42 kV
Tank Hole Diameter	=114.3 mm
Porcelain Color	=gray

				15kV/1227A BIL 110kV			EVER POWER			
Design	JULY LI	Check							Transformer Bushing	
Approve		Craft			Weight	Scale	DWG NO			
		Date	2024.03.06			1:1	GN 1724-001 UE			



Exposed copper and brass parts tinned, thickness >5 um

Rated Voltage	=46 kV ✓
Rated Current	✓ =417 A
Creep Distance	>=1322 mm ✓
Impulse Level	=250 kV ✓
dry withstand Voltage	=95 kV ✓
Tank Hole Diameter	=108 mm
Porcelain Color	=gray

ITEM	DESCRIPTION	QTY	MATERIAL
1	GNV 7060 46kV 417A BIL 250kV CD=1320 Porcelain Bushing	1	Porcelain
2	Plain gasket Ø82.5XØ50.8X7.2-J	1	NBR
3	Brass Cap 46kV(417A)-J	1	Brass Forging
4	46kV 417A Copper Stem (CD=1320)	1	Copper
5	End cap gasket 417A-J	1	Silicone rubber
6	Pass oil ring (417A)-J	1	Aluminium alloy
7	Brass nut 5/8"-11	4	Brass
8	Brass washer Ø30XØX17X2.5	2	Brass
9	Gasket Ø150XØ118X3.3	1	NBR
10	Flange(Ø215.5-17.5)	1	Aluminium alloy
11	Flange gasket(Ø155X122X6)	1	NBR
12	Tube	1	Mylar Tube

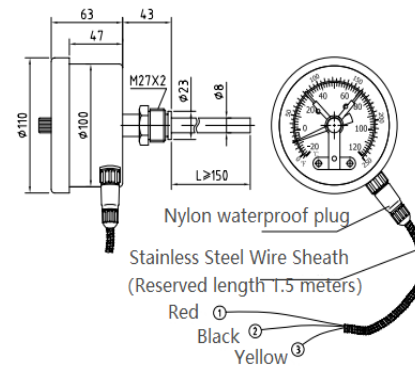
					HV Bushing		EVER POWER	
					46kV/417A ✓ BIL:250kV		Transformer Bushing	
Design	JULY LI	Check					DWG NO	
Approve		Craft				Weight	Scale	
		Date	2024.04.10				1:1	GN1771-006 UE

Oil Temperature Indicator

BWPK-802A-2 transformer thermometer is a special instrument for temperature measurement and control of various small transformers. The instrument has the characteristics of sensitive response, clear indication, convenient installation and good reliability. When the instrument reaches the set temperature, it can output an electrical signal to measure and control the temperature of the transformer in a remote room. The instrument adopts stainless steel shell, beautiful appearance, good protection performance, and can be used outdoors for a long time.

Dimensions

2. Dimensions

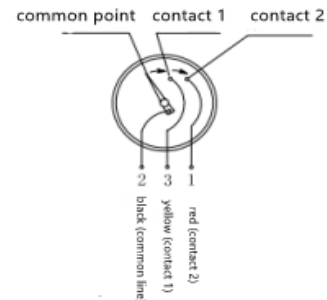


Technical data

1. Measuring range: -20~120°C (0-100°C, 0~150°C)
2. Accuracy: +/-1.5
3. Structure type: Axial
4. Mounting thread: M27X2
5. Bulb sensor dimension: Φ8X150
6. Connection form: Two contacts
7. Connection capacity: 10VA/220V (resistive)

Wiring diagram

4. wiring diagram



NOTICE
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MODEL
NO.

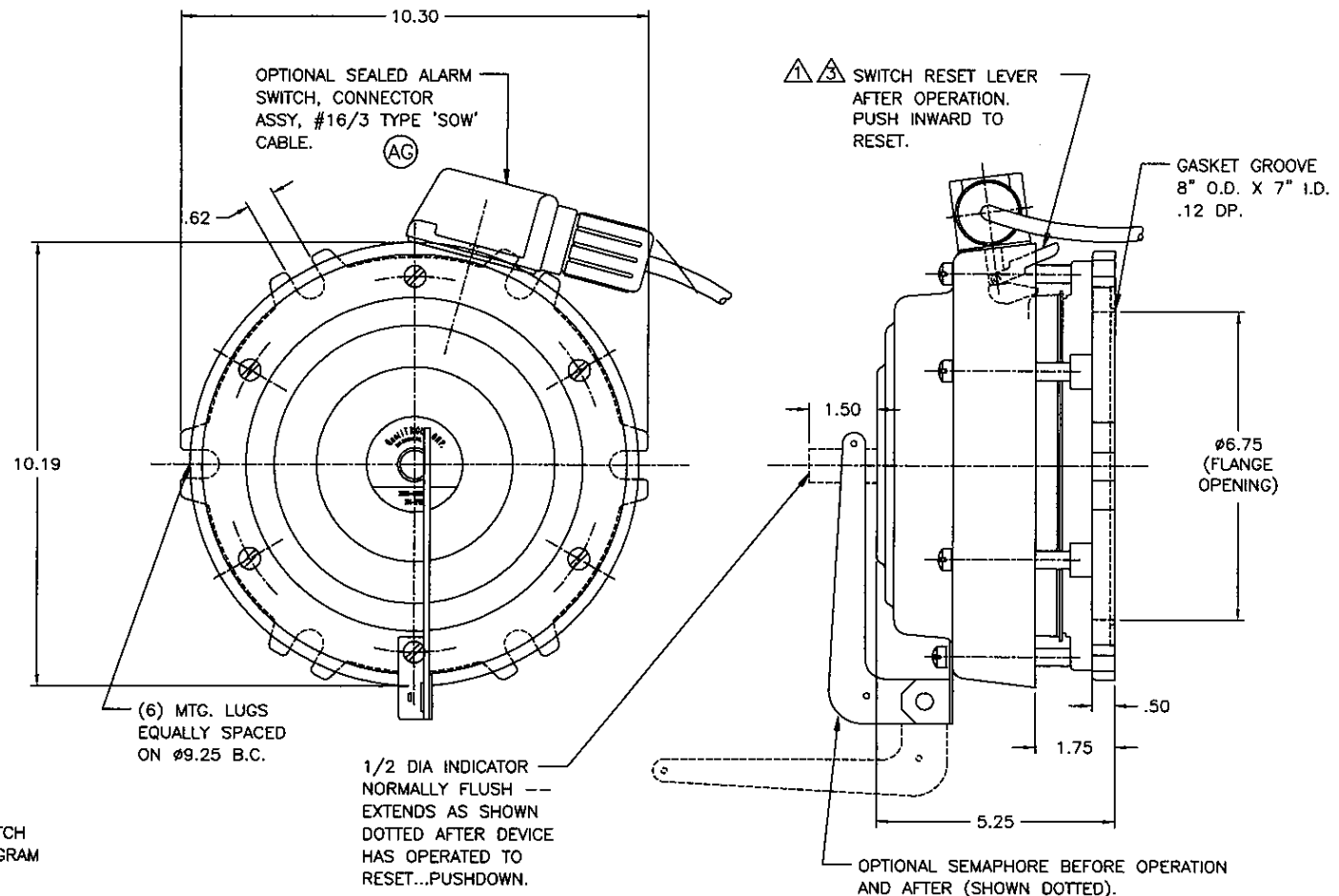
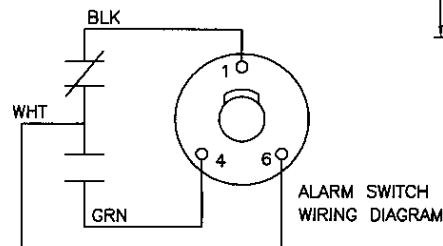
208-60

NOTES:

1. BLACK AND WHITE ARE NORMALLY CLOSED WHEN ALARM SWITCH IS MOUNTED AND ARMED AS SHOWN.
2. WHEN DEVICE OPERATES: BLACK AND WHITE ARE OPEN GREEN AND WHITE ARE CLOSED
3. WHEN THE SWITCH RESET LEVER IS MAN. DEPRESSED: BLACK AND WHITE ARE CLOSED GREEN AND WHITE ARE OPEN
4. FINISH: ASA #70 LT GRAY EPOXY

SWITCH RATING

15 AMP @ 125,250,480 VAC
.5 AMP @ 125 VDC
.25 AMP @ 250 VDC } NON-INDUCTIVE



SHEET 1 OF 3

4-6-04	ADDED "OPTIONAL" TO SEMAPHORE CALLOUT	24381
06-08-98	DELETED PPS-65 NOTE; 1.50 WAS 1.81; UPDATED A-SIZE FORMAT	19508
LET	DATE	ALTERATION
		C.N.

QUALITROL
CORPORATION
DIMENSIONS AND TOLERANCES IN
ACCORDANCE WITH ASME Y14.5M-1994

PH. (716) 586-1515
FAX (716) 377-0220
1385 FAIRPORT ROAD
FAIRPORT, NY 14450
FORM 0-473 REV.C 7/1/97

TITLE: PRESSURE RELIEF DEVICE
MODEL NO. 208-60

速动油压继电器

SYJ₉Quick-acting oil pressure relay

一、产品介绍

SYJ₉ 速动油压继电器是沈阳科奇电器有限公司新研制的用于保护变压器、电力电容器、电抗器等油箱安全的一种压力保护继电器，它以油箱内部压力变化速度作为测量信号。当油箱内压力升高速度大于 2~6KPa/sec 时，速动油压继电器即可动作，其动作响应时间随着压力梯角的升高而缩短，其动作特性曲线参见（图 1）；SYJ₉ 速动油压继电器动作时其压力的净增加值，即升高速度与响应时间的乘积，小于等于 25KPa。例如压力以 500 KPa/sec 梯度上升时，其动作响应时间小于 0.05 秒（参见图 1）。本继电器的动作特性可使变压器在突发恶性事故时油箱压力处于低压力区即可报警，切断电源，同以往的常规气体继电器相比性能可靠，动作灵敏，可大幅度减少变压器因故障所造成的损失。使用本产品时，推荐 SYJ₉ 速动油压继电器信号跳闸。

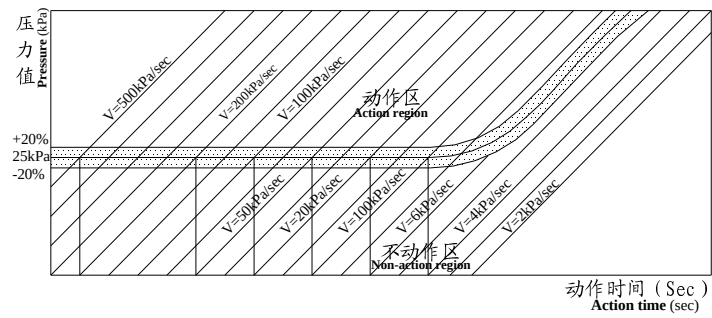
本产品还可实现双路电气信号输出，一路接跳闸，另一路接计算机系统。

Introduction of the product

SYJ₉Quick-acting oil pressure relay is a new type of pressure protection relay for protecting the safety of oil tank of transformers, power capacitors, reactors, which is designed by Shenyang Keqi Electrical Equipment Co., Ltd.. It measures according to changing velocity of internal pressure of fuel tank. When the rising speed of pressure in fuel tank exceeds 2~6kPa/sec, the quick-acting oil pressure relay will start up. Responding time will be shortened along with the rise of pressure and its characteristic curve of action, see Figure 1. When acting, the added value of pressure, that is, rising speed multiplies responding time, shall be not more than 25kPa. For example, when the pressure rises at the grade of 500kPa/sec, its action response time shall be less than 0.05 second, and when the pressure rises at the grade of 100kPa/sec, its action response time shall be less than 0.25 second (See Figure. 1). The characteristic of this relay is to facilitate the transformer to keep fuel tank pressure in low pressure region and give an alarm and cut off power in case of malignant accident. Compared to previous regular gas relay, it is characterized by reliability and quickness, which can greatly reduce the loss causes by the malfunction of the transformer. When using this product, it is recommended that SYJ₉Quick-acting oil pressure relay connected to the trip.

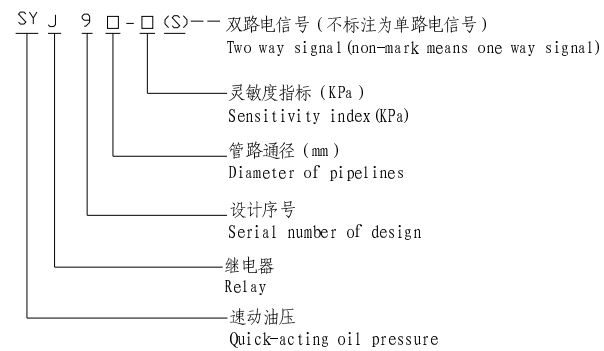
This product can realize to send out electrical signal in two ways, one shall be connected to trip and the other shall be connected to computer system.

—图 1—
—Figure 1—



二、型号及意义

Symbol and its meaning



标注示例:

SYJ₉-50-25 (S) 表示与变压器或电抗器相连的所配蝶阀通油口径为 $\Phi 50\text{mm}$, 灵敏度为 25KPa, 双路电信号的速动油压继电器。

For example:

SYJ₉-50-25 (S) indicates that butterfly valve diameter matched with transformer and reactor is $\Phi 50$ mm, sensitivity index is 25kPa, two-way signal quick-acting oil pressure relay.

三、工作原理

本产品分为油室和电器室两部分，由一隔离板隔开，在油室内设有一个大隔离波纹管，大波纹管与微力波纹管联通，组合成一个密封腔体，该密封腔体与电器室通过一个微管平衡器相连通，大波纹管内设平衡弹簧，该部组成 SYJ₉ 速动油压继电器的压力梯度与位移转换部分，即当油压突然增高时，密封腔收缩，内部压力增高，推动微力波纹管伸长，该伸长位移推动微动开关动作，发出报警信号，当压力缓慢增高时，密封腔中空气压力缓慢增加，该压力增高经微管平衡器泄压至电器室，致使微力波纹管动作减缓甚至于不产生位移这样就达到了只有当压力升高梯度达到一定值以后，才能动作这一结果，且压力升高梯度越大，动作响应时间越短。

Working Principle

This product divided into two parts with a separated plate, one is oil cavity and the other is electric room. There is a big isolating corrugated tube in the oil cavity, and the big corrugated tube connected with micro force corrugated tube, and they are composed a seal cavity. This cavity is connected with an electric room through microtubule balancer. There is a balance spring in big corrugated tube; this is the part that composes the pressure gradient and displacement transfer of the SYJ₉ Quick-acting oil pressure relay. It is that when the oil-pressure increased abruptly, the seal cavity will shrink, and the internal pressure is increased, it will impel the elongation of the slight pressure corrugated tube, and the elongation of the displacement will impel the action of the micro switch, and it will send out an alarm signal. When the pressure increased slowly, the air pressure that in the seal cavity is increased slowly, the increased pressure is released to the electric room through microtubule balancer. It makes the action of slight pressure corrugated tube decreased slower, or so much as did not have displacement. Only the pressure increase gradient reaches to the certain value, it can get result of the action. And the more the pressure increase gradient, the shorter of the time that the action affect.

四、使用范围及技术参数

Scope of use and the technical parameter

1. 使用范围：

- (1) 适用于油浸式变压器、电抗器及其它封闭的油浸电器室；
- (2) 环境要求：-30℃~+50℃；
- (3) 相对湿度：+20℃时，不大于 95%。

Scope of use

- (1) It is indicated for the oil-immersed transformer, reactor and other airtight oil-immersed electric room;
- (2) Environmental requirement: -30℃~+50℃;
- (3) Relative humidity: when temperature is +20℃, it is no more than 95%;

2. 技术参数:

- (1) 信号开关电器性能参数见 (表 2)
- (2) 电气绝缘性能: 微动开关接线端子间及导电部分对地应承受工频电压 2000V, 持续 1min, 无绝缘体发热, 击穿, 闪络现象;
- (3) 抗振动能力, 在振动频率 4~20Hz (正旋波), 加速度 2~4g 时, 继电器接点不动作;
- (4) 外壳防护等级符合 GB4208 标准中 IP55。

Technical parameter

- (1) The electricity performance parameter of signal switch (see Table 1)
- (2) The electrical insulation properties: among the micro switch terminal themselves or the current carrying part to the ground should be endure the line frequency voltage for 2000v, and it should be persistence for 1 min, and the insulator does not have the phenomena of give off heat, puncture, scintillation;
- (3) Anti vibration ability: when the vibration frequency is 4—20Hz(sine wave) and the acceleration is 2—4 g, the tip do not action.
- (4) Level of shell protection is accord with IP55 in the standard of GB4208.

—表 1—
—Table1—

种类电压 Category voltage	电压 (V) Voltage	电阻负载 (A) Resistance Load	感性负载 (A) Inductive Load
AC	220	5	3 (cos Φ=0.4)
DC	220	1	0.1(T=7ms)

五、外型尺寸 (参见图 2)

External dimension (see figure 2)

六、结构及特点

Structure and Features

1. 结构

继电器外壳是由高强度铝合金型材通过机械加工制成, 内部安装有隔离波纹管, 把电器室和油室隔开, 微动开关和平衡器都安装在电器室内, 本产品油室部分对外设有放气塞及排污口, (放气塞同时作为现场定期检测口, 见图 2), 放气塞安装时开口向上或水平放置。

Structure:

The shell of relay is made of high strength aluminum alloy section material through the machine work. Isolating corrugated tube is set up in the inner side; it separated the electric room and the oil cavity. The micro switch and the balancer are set up in the electric room. The oil cavity of this product has air relief cock and waste outlet, (the air relief cock also as detection interface on site regularity. See figure 2), when you set up the air relief cock, the placket should be placed upward or horizon.

2. 特点:

- (1) 本产品采用合金铝型材加工而成，没有渗漏、断裂的隐患；
- (2) 产品的安装位置与最高油面的垂直距离可选择位于 0~6m 之间；
- (3) 工作特性精确可靠；
- (4) 本产品配备检测装置，例行在线测试方便；可提供双路电气报警系统，便于计算机监控管理。

Features

- (1) This product is made of alloy aluminum section, and has no hidden trouble of leakage and rupture.
- (2) The vertical distance of the products between the mounted position and tiptop oil level can be choose to located among 0—6 m
- (3) Operation factor is precision and true
- (4) This product provides the detector set, and the routine online examination is convenient; May provide the two-circuit electricity alarm system, and it is advantageous for the computer monitoring management.

七、SYJ₉ 速动油压继电器外接线推荐接线（见图 3）

VII. The recommend connection diagram of external wire of SYJ₉

Quick-acting oil pressure relay. (See figure 3)

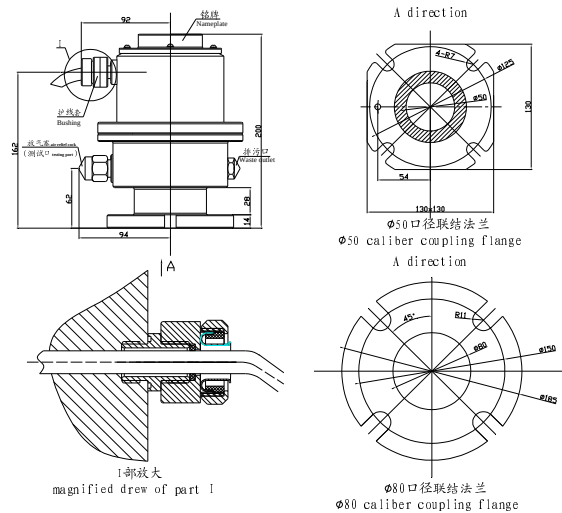
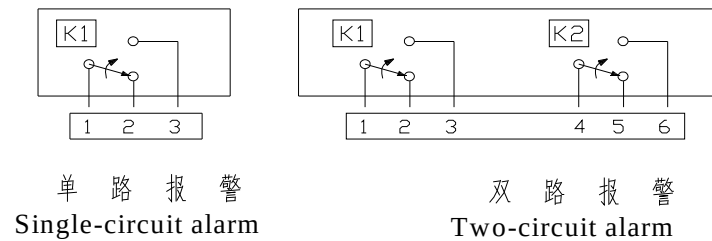


图 2 速动油压继电器外形尺寸图

Figure2.The overall dimension drawing of quick-acting oil pressure relay

—图 3—
—Figure3—



注：图中 K1、K2 为各自独立的报警用无源开关，K1 接至端子 1、2、3；K2 接至端子 4、5、6。

K1：1、2 间为常闭接点用于指示正常，1、3 间为常开接点用于报警或跳闸报警；

K2：4、5 间为常闭接点用于指示正常，4、6 间为常开接点用于报警或跳闸报警。

Note: In the figure, K1 and K2 are individual swifts of alarm power supply, K1 connects to the posts 1, 2, and 3; K2 connects to the posts 4, 5, and 6.

K1: it is normally closed contact between 1 and 2, which is used for indicating normal condition, and it is normally open contact between 1 and 3, which is used for alarm or tripping alarm;

K2: it is normally closed contact between 4 and 5, which is used for indicating normal condition, and it is normally open contact between 4 and 6, which is used for alarm or tripping alarm.

八、与变压器的连接方式、安装及维护

- 1、在变压器运输过程中，禁止把 SYJ₉ 速动油压继电器安装在变压器上，并且不要使继电器受到撞击；
- 2、SYJ₉ 速动油压继电器通过一个 $\Phi 50\text{mm}$ 或 $\Phi 80\text{mm}$ 的蝶阀安装在变压器油箱侧壁上或箱盖上，其安装位置距储油柜中油面的垂直距离为 H，如图 5 所示，当 H 大于 3m 时，定货时请特别注明；
- 3、当变压器为强油循环冷却时，继电器不应安装在靠近出油管口的正对区域，以免在运行和停止油泵时，继电器出现误动作；
- 4、SYJ₉ 速动油压继电器最好安装于油箱侧壁（也可安装于油箱盖上），放气塞在上端（联结法兰的尺寸参见图 2）
- 5、SYJ₉ 速动油压继电器安装完毕后，接通信号线路，并将与之相联的蝶阀打开；
- 6、SYJ₉ 速动油压继电器被正确安装后，蝶阀处于打开状态，卸下放气塞上的防护帽，将测试头的透明连管一端从打气筒外接头中卸下，另一端插入测试帽快速接插头内并将测试帽顺时针旋入放气塞，直至少量油流出，将测试帽卸

- 下，擦净出油口，安好防护帽（参见图 4）；
- 7、外接引线时卸下顶部防雨盖，卸下产品出厂所测试引线，将正式引线按防雨盖上接线图接至端子，拧紧出线口锁紧螺母，使密封胶圈将引线压牢，安装好防鱼盖（厂方需要可在外接引线外安装护线金属软管，本产品紧锁螺母后部配有金属软管接头）。
 - 8、产品定期检验时，可将蝶阀关闭，用专用检测装置通过放气塞对报警性能进行简单试验，也可将继电器拆下后，进行性能试验，合格后再安装到变压器上，如继电器性能不能符合标准要求，可将备品换上，不合格品返厂维修。
（注：我公司可根据拥护需求，生产制造本继电器检验装置，如用户需要可向我公司订制）。
 - 9、如需提取油样可将导油管与放气塞打开，提取油样后，将放气塞关闭即可。
- 注：**SYJ₉ 速动油压继电器在出厂前已整定合格，除正常接引出线外，用户严禁拆开产品调试继电器报警系统。

Way to connect install and maintenance with transformer

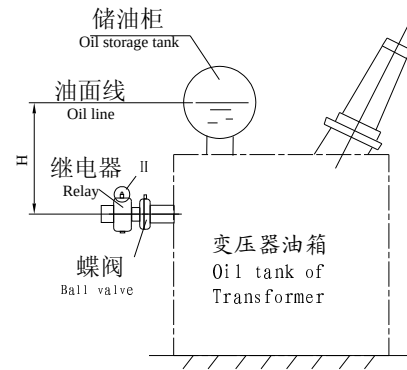
1. During transformer transportation, it does not allow installing SYJ₉ quick-acting oil pressure relay on the transformer, and do not let the relay be hit;
2. The SYJ₉ quick-acting oil pressure relay must be installed through a butterfly valve of Φ50mm or Φ80mm on the sidewall of transformer oil tank or the case lid, the vertical distance from its installation position to the oil level in the oil storage tank is H, as shown in Figure 4, if H is more than 3m, please give a special indication when ordering;
3. When transformer for Forced-Oil-Air circulation cooling, the relay should not be installed approaching the facing area of oil pipe end, in order to avoid mal-operation of the relay in running and at stop of oil pump.
4. it is advisable that install SYJ₉ quick-acting oil pressure relay levelly with the sidewall of oil tank (also may install on the oil tank cover), Air relief cock is in the upper (size for flange joining see Figure 2) .
5. When finished the installment of SYJ₉ quick-acting oil pressure relay, connect signal line, and open the butterfly valve which connected with it;
6. After SYJ₉ quick-acting oil pressure relay is installed correctly, the butterfly valve being at its open condition, dismounts the protective cap from air relief cock, dismount the transparent one end of connecting pipe of test joint from the air pump external connection, put another end insert the quick connection plug of test cap and clockwise turn the test cap into air relief cock, until the few oil streams out, dismount the test cap, scours the output port, then well install protective cap (see figure 4)
7. when connecting lead wire from outside, dismount the crown rainproof lid, dismantle test lead wire that attached with the product according to the wiring diagram of rainproof covers, connect the official lead wire to the post, Screw tight the locking nut at cable outlet, and let the rubber seal press tight with lead wire. Install the rainproof lid (If the factory needs, it is possible to install wire metal hose outside the lead wire from outside, Behind this product locking nut there is a

metal hose attached) .

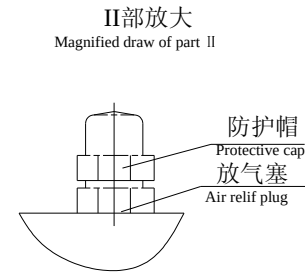
8. When carrying out product inspection ordinary, may close the butterfly valve, use special detection device through the air relief cock to carry out the simple test about alarm performance. The performance test also can be carried out after dismantle the relay. After qualified, install on the transformer again, If performance of the relay fail to conform to the standard, May exchange the spare parts, and the defective products will be returned back the factory for maintenance service. (note: Our company can product and manufacture this checking device for relay as users' demand, users who need can make an order to our company) .
9. If there is need to extract oil sample, connect the oil lead with the air relief cock, turn on the air relief cock, after extract oil sample, turn off the air relief cock then.

Note: The SYJ₉ quick-acting oil pressure relay has been qualified before leaving the factory, Besides receives and instructs the going beyond a line normally, the user must not disassemble the product to debug relay alarm system.

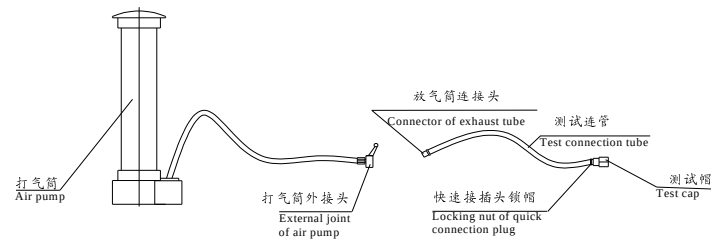
—图 5—
—Figure5—



—图 4—
—Figure4—



—图 6—
—Figure6—



九、继电器的检测方法

对继电器进行检测时，可分为例行检测及性能测试两种。

1. 在线检测可运用我公司随产品配备的检测装置（见图6），将
 - （1）关闭继电器与变压器之间的蝶阀，连好外测试电气线路；
 - （2）将实验用打气筒外接头与放气管接头连通；
 - （3）卸下放气塞防护帽，将测试帽旋入继电器放气塞接口，使放气塞打开；
 - （4）如安装高度为 3m 以下，使打气筒快速打气 1~2 次；如安装高度为 3~6m 之间，用打气筒快速打气 3~4 次，观察继电器是否动作，动作视为正常，不动作视为电气报警失灵。
2. 性能测试，具体步骤如下：
 - （1）关闭继电器与变压器之间的蝶阀，连好外测试电气线路；
 - （2）将继电器安装到检测台上对继电器进行全面检测，等同出厂试验，如继电器检测不合格可换上新的继电器，打开蝶阀，打开放气塞使继电器充满油后关闭放气塞，安好防护帽，不合格品返厂维修。

Method for relay examination

It can be divided into two kinds to carry on the examination to relay: the routine online examination and the performance test.

1. To carry on the online examination, you may use our detection device which provides along with the product (see Figure 6), connect the test cap and the air relief cock to carry on the experiment. The concrete step is as follows:
 - (1) Close butterfly valve which is between the relay and the transformer, connect well the electrical line of outside tests;
 - (2) Connect the external joint of experimental air pump and the discharge pipe joint;
 - (3) Dismount protective cap of the air relief cock, turn the test cap into the interface of relay air relief cock, and open the air relief cock;
 - (4) If installation height is below 3 meters, use air pump to inflate once or twice fast; If installation height is between 3 meters and 6 meters, use air pump to inflate 3-4 time fast, observe the relay whether moves or not, if it moves, that means it is in

normal condition, and if it fail to move, means the electricity alarm fails to operate.

2. As to performance test, the concrete step is as follows:

(1) Closes butterfly valve, clean out the transformer oil from the waste outlet, then dismount the relay;

(2) Install the relay on test table, carries on all-around test to the relay, same as examination before leaving the factory. If the relay unqualified, exchange a new relay, turn on reed valve, after turn on the air relief cock to fill up the relay with oil then close the air relief cock, fit well protective cap, The defective products will be returned back the factory for examination and maintenance.

十、订货须知

用户订货时须提供以下内容:

- 1、所订产品名称、型号、配用蝶阀的口径;
- 2、产品安装位置超过 3m 时, 需提供其安装位置尺寸;
- 3、海拔超过 200m 时请注明;
- 4、产品数量、供货期;
- 5、联系方式、联系人及交货地点;
- 6、所需产品、备件说明;
- 7、特殊环境

Notice for order

Following content should be provided when make an order:

1. name of product, model, caliber of matched butterfly valve;
2. if installation position of the product exceeds 3 M, please let me know its dimension installation position;
3. Please indicate when the elevation surpasses 2000m;
4. Product quantity, Supply time;
5. Connect method, connect person and delivery point;
6. Needed spare parts, spare parts explanation
7. The special environment and the special request explanation of installation and application.

十一、质量控制

本继电器在制造过程中严格执行质量控制程序, 每台继电器都根据相关标准的要求, 经过如下项目的检验: 继电器的结构和外观; 密封试验; 报警开关接点间及其对地工频耐压试验; 电器室防雨试验; 突发压力动作性能试验等。

Quality control

The relay executes strictly the quality control procedure when it is made. Each relay is conforming the relative standers, through the following item verify: the Structure and appearance of the relay; seal test; electric strength test; rainfall test; outburst press action performance test and so on.

十二、承诺

用户单位在使用本产品时要按说明书进行操作，本产品在安装之日起 18 个月内，并在公司发货给用户不超过 24 个月的期限内，我公司对产品实行三包。

Promise

When the unit of user uses this product, it must carry on the operation according to the instruction booklet. Our company carries out three guarantees to the product, when it is in the date of 18 months in installment, and delivers goods in the company for the user does not surpass for 24 month-long in the deadlines.

This instruction booklet is pressed in July.2006.

If there is any mend on technical parameter, it is subject to the new instruction booklet

No special information for it, if there is any dissent please contact with our company

BWR—04 Series

***Winding Transformer
Temperature Indicator***

Operation

Instructions

Hangzhou HuaLi Instrument & Meter Co.,Ltd

Address:27 Longwu Industrial Zone, Hangzhou Post code:310024

Tel : 0571-87088120

E-mail: lanshan@mail.hz.zj.cn

Fax: 0571-87919920

Http://www.hz-hlyb.com

Thanks for using our product
Before operating it, please read this manual thoroughly

Contents

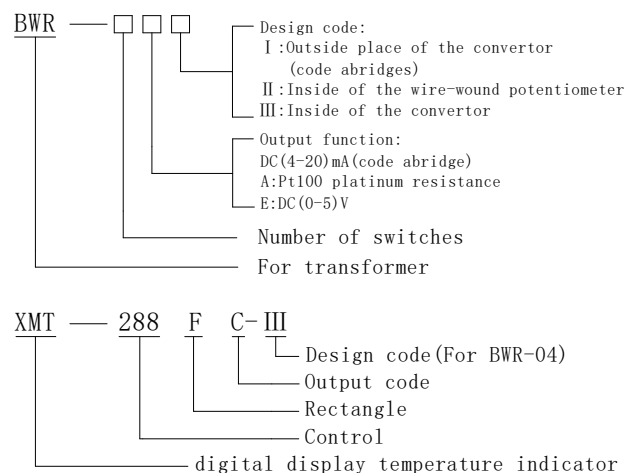
1.Brief Introduction	(1)
2.Model Illuminate.....	(1)
3.Composing And Function	(1)
4.Main Technical Parameters.....	(4)
5.Installation	(5)
6.Wiring.....	(6)
7. Setting & Adjusting	(9)
8.Points of Attention.....	(12)
9.Addendum	(13)

1. Brief Introduction

Transformer winding thermoregulator is specially designed for oil-immersed transformer, which can indirectly measure the winding temperature by thermal simulation method.

Let T_1 present the temperature of transformer winding and it is the sum of T_2 and ΔT , i.e. $T_1 = T_2 + \Delta T$, where, T_2 is the oil temperature in upper part of the transformer, and ΔT is the winding temperature rise relative to the oil temperature. The ΔT depends on the winding current and the secondary current of the current transformer (CT) is in proportion to the winding current. The thermoregulator works in following. Principle: The current from the CT, which is in proportion to the load, is first adjusted via a converter, and then flows through the heating element embedded in the elastic component; the heat produced by the heating element leads to an additional displacement of the elastic component, and thus a temperature indicated value higher than the oil temperature. By this way, the thermoregulator can obtain an average indicated value of the winding temperature.

2. Model Illustrate



The model and illuminate of the transformer winding temperature indicator:

3. Composing And Function

1. BWR-04 series winding temperature indicator:

- (1) Indicate the temperature of transformer winding.
- (2) Afford four sets of switch (AC250V/15A), which can be set within a whole span, are

employed to respectively control over the transformer cooling system, overheating alarming and tripping, etc.

(3) Output (4~20)mA electricity signal to be the input of the digital display thermometer which is in the controlling room.

The outline drawing and mounting dimension of the winding temperature indicator refer Figure 1. The outline drawing and mounting dimension of the export of BWR-04 refer addendum (offer the overseas transformer producer)

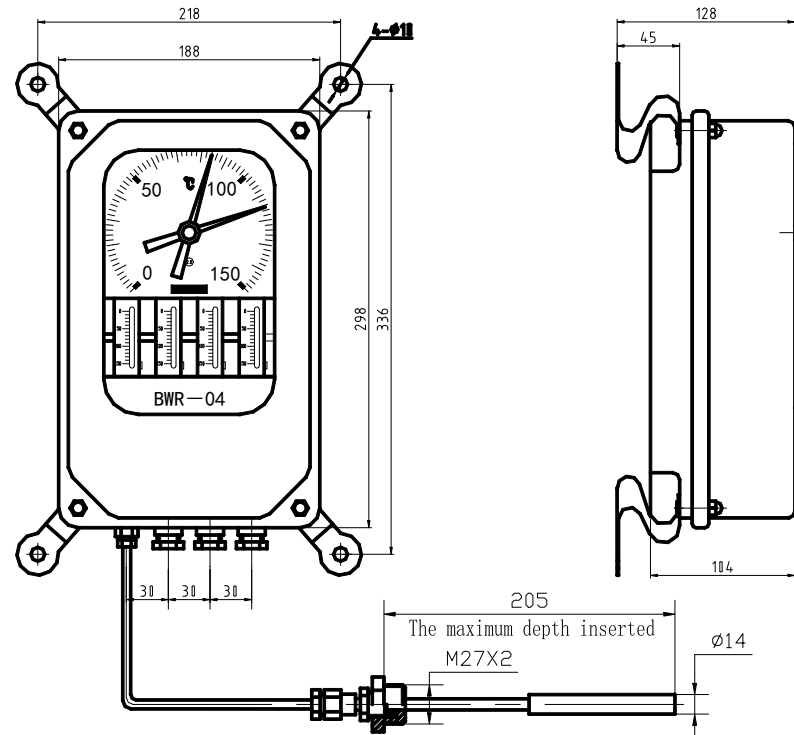


Figure 1 BWR-04 thermoregulator outline drawing and mounting dimension

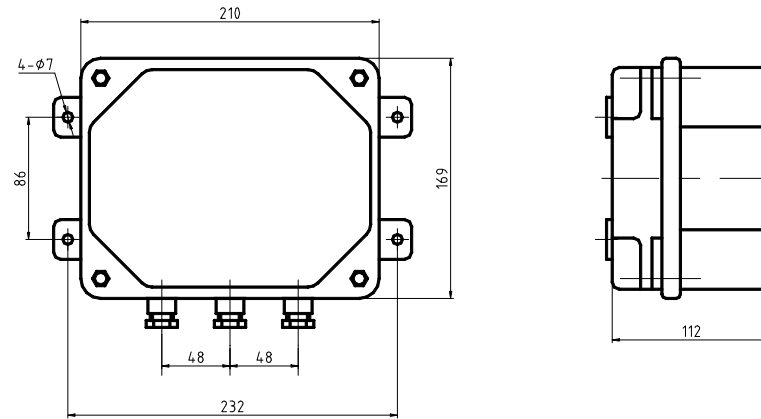
2. Convertor

The function of the convertor is got electricity for the CT of the main transformer. Then after

adjust, send it to the heating element of the winding temperature indicator. So that the indicator can show a temperature value which is high then the oil temperature which is ΔT .

The convertor that bases on the different rating electricity, which is 5A, 3A, 2A, 1A, of CT of the main transformer, which can be divide into four different connection. Idiographic connection method refers table 2 or the connection diagram in the convertor cover .

The outline drawing and mounting dimension of the convertor refer Figure 2



3.XMT—

The transformer change the temperature into the (4~20) mA electricity signal, then send into the digital display thermometer XMT—288FC—III, which is fixed in the controlling room. The man who is on duty can expediently observe the temperature of the transformer in controlling room. And it also can send the signals whatever you need for computer as follows:

- (1)(4-20) mA direct current signal.
- (2)(0-5)V or (1-5)V constant voltage signal.
- (3)RS232, RS485 communication interface.

The outline drawing and mounting dimension of XMT—288FC—III refer Figure 3.

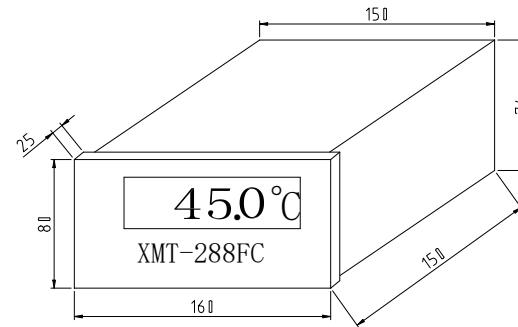


Figure 3 XMT—288FC—III outline drawing and mounting dimension

4. Main Technical Parameters

1.BWR-04 Main Technical Parameters:

- (1) NORMAL WORKING CONDITION: $(-40\sim+55)^{\circ}\text{C}$
- (2)Range of measurement: $(0\sim150)^{\circ}\text{C}$
- (3)Indication accuracy: $\pm 3^{\circ}\text{C}$
- (4)Controlling capability:
 - ①Setting range: adjustable within the work span ②Setting accuracy: $\pm 3^{\circ}\text{C}$
 - ③switch difference: $6\pm 2^{\circ}\text{C}$ ④RATED POWER: AC250V/15A
- (5)standard setting point:
 $K_1=55^{\circ}\text{C}$ $K_2=75^{\circ}\text{C}$ $K_3=90^{\circ}\text{C}$ $K_4=130^{\circ}\text{C}$
- (6)Protection performance: IP55

2.Convertor Main Technical Parameters:

- (1) NORMAL WORKING CONDITION: $(-40\sim55)^{\circ}\text{C}$
- (2) Protection performance: IP55
- (3) Main technical parameters refer table 1.

3.XMT-288FC-III Main Technical Parameters:

- 1. Normal working condition: $(-10\sim50)^{\circ}\text{C}$
- 2. Measuring range: $(0\sim150)^{\circ}\text{C}$
- 3.Input signal: DC (4-20) mA
- 4. Accuracy: 1.0
- 5. Resolution ratio: 0.1°C
- 6.Controlling capability
 - (1) Controlling mode: Upper limit contact
 - (2)Setting range : Adjustable within the work span
 - (3) Contact capacity: AC220V, 3A
- 7.Output: DC(4-20)mA

Notice: It can be made with RS232,RS485 or other output signal according to customer's order.

TABLE 1 Converter Technical Parameters			
Type	Input current I _P (A)	Output current I _s /Input current I _p	Adjust lead K connect to port's NO.
A	$5 \geq I_p > 3$	0.30~0.35	17-6
		0.24~0.30	17-7
		0.15~0.24	17-8
		0.10~0.15	17-9
B	$3 \geq I_p > 2$	0.50~0.60	17-7
		0.40~0.50	17-8
		0.28~0.40	17-10
		0.17~0.28	17-11
C	$2 \geq I_p > 1$	0.75~0.90	17-12
		0.60~0.75	17-13
		0.40~0.60	17-14
		0.25~0.40	17-15
D	$1 \geq I_p > 0.61$	1.50~1.75	17-6
		1.20~1.50	17-7
		1.00~1.20	17-14
		0.50~1.00	17-16

5. Installation

1. Installation of BWR-04 and Converter

Screw 4 hexagonal bolts (M8) and 4 matched washers accessorized ($\phi 10$)with BWR—04 into a mounting board of transformer.

Screw 4 hexagonal bolts (M6) and 4 matched washers accessorized ($\phi 7$) with Converter into a mounting board of transformer.

2. Installation of bulb

Wrap the installation screw thread on the bulb with appropriate seal material. Insert the bulb into the transformer's sleeve with oil inside it. Screw the mounting screw on the bulb into the sleeve until fasten the sealing washer. At last screw the airproof bolt fast.

3. Installation of XMT-288FC-III

The XMT-288FC-III install on either an operations area or an instrument screen with two mounting racks and two screws which provided. The size of aperture on instrument screen is high* wide =(152*76) mm.

6. Wiring

1.Connection of convertor

You have to take off the cover to wire the adaptor convertor, loose the screws which is on the lead terminal, take out the airproof in the lead terminal , and then you can connection the lead.

Depend on the difference of the maximum current of CT, the method of the convertor has A,B,C,D four kinds. The connection diagrams of convertor refer to table 2.

TABLE 2 The Connection of Convertor		
Type	The Maximum	Connection Diagrams (R: Heating Element K: Adjusting Lead)
A	5.0A	
B	3.0A	
C	2.0A	

Figure 4 General connection of BWR-04/convertor / XMT-288FC-III

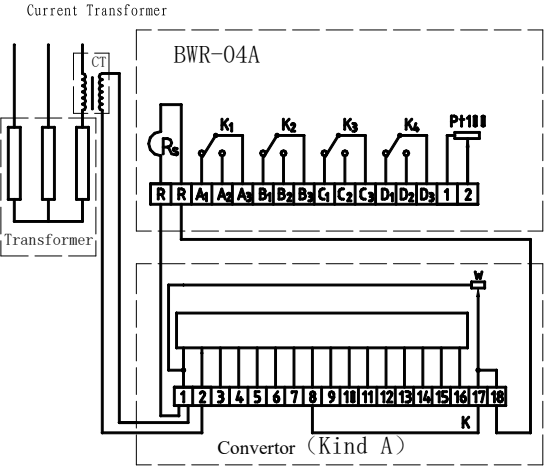


Figure 5 BWR-04A/Convertor connecting diagram

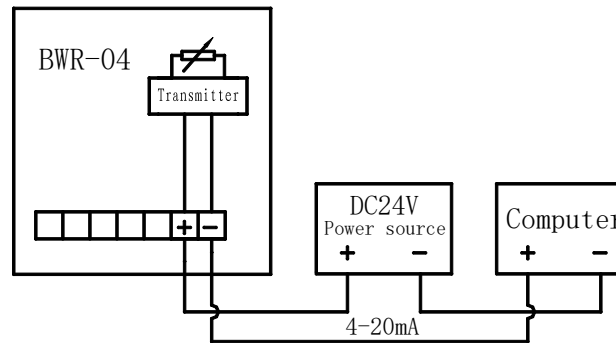


Figure 6 BWR-04/DC24V power source/Computer connecting diagram

7.Setting & Adjusting

1. Setting of switch of BWR-04

Each switch can be set individual. The temperature control switch K_1, K_2, K_3, K_4 of BWR—04 have been already set at 55°C , 75°C , 90°C , 130°C before its delivery. If users need to change the value of set points, please follow below procedures:

- (1) Loose the round bolt at the point of the red arrowhead;
- (2) Stir the red setting needles to set value planned;
- (3) Screw down the round bolt;

(4) Circumgyrate the axes of the scale in a clockwise direction (suppose watch the axes form the left direction). The value of the indicator will go form lower to higher. Check the value of the indicator when the switch moves.

Notice: The axes will never circumgyrate the opposite direction; otherwise, it may change the accuracy of the indicator.

2. Setting of convertor

Before the convertor delivery, the input current of the heating element had been set at 1.04A, depending on the Maximum Current of CT which is 5A and the average rising temperature of the transformer winding which is $\Delta T=20K$.

When the condition not accord with the parameter above, the convertor can be reset before use. The adjusting connection refer Figure 7a. the instrument which used in the figure as follows:

If the customer has no AC electric source, can adjust to press the 2KVA regulating transformer and 1.2~1.5KW heating. The connecting diagram refer Figure 7b. The instrument which used in the figure as follows:

- (1)B: 2KVA/250V Regulating transformer
- (2) R_L : Heating of 1.2KW~1.5KW
- (3) A_1 : Model D26-A current meter of grade 0.5 for (0-5)A
- (4) A_2 : Model D26-A current meter of grade 0.5 for (0-2)A

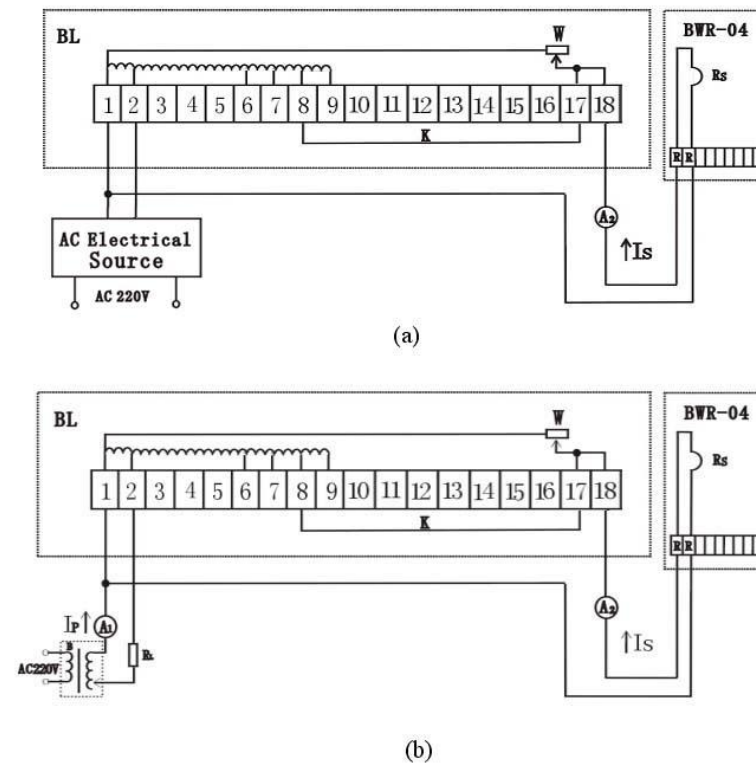


Figure 7 Adjusting connection of the convertor

Adjusting step as follows:

(1) According to the rating current of the transformer and the no-load voltage ratio of current transformer calculate the output current of the current transformer.

For example:

the rating current of the transformer: 400A

the no-load voltage ratio of the CT 400:5

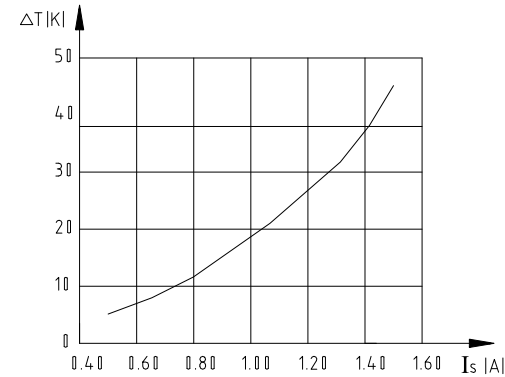
$I_p = 400 / (400/5) = 5A$

(2) Fine out the average rising temperature (ΔT) from the transformer manual or get it from the producer of the transformer.

For example, $\Delta T = 20K$

(3) According to the value of the ΔT , get the value of I_s from Figure 8. For example,

According to $\Delta T = 20K$, get $I_s = 1.04A$



$\Delta T(K)$	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38
$I_h(A)$	0.74	0.80	0.86	0.92	0.98	1.04	1.09	1.14	1.19	1.24	1.28	1.32	1.63	1.40	1.44

Notice: If the value you want is not in the table, you can get it in the diagram above.

Figure 8 Temperature rising characteristics of heating element

(4) Calculate I_s/I_p , after according to the value of I_p and I_s/I_p , depending on Table 1, It will be that the adjusting lead K is the wiring point. For example $I_s/I_p = 1.04/5 = 0.208$. Depending on Table 1, K from No.17 to No.8.

(5) Connect which as figure 7 shows.

(6) Turn on the power, initialize the output current of the AC electrical source equal to the rating output current I_p of the current transformer (Figure 7a) or adjust the regulating transformer B to make the output current (the value of A_1) equal to the rating output current I_p of the CT (Figure 7b). For example, $I_p = 5A$.

(7) Adjust the rheostat in the convertor (W), make $I_s = 1.04A$ (the value of A_2).

8. Points of Attention

1. When checking BWR-04, the bulb must be immersed into constant temperature trough entirely, the indicator must be fixed perpendicularly.

2. When fixing the BWR-04 or wiring, must be careful and gently. Please follow the step of the manual strictly.

3. Make use of the solid wrench to open and close the meter cover. In particular, while closing the cover, screw 4 hexagonal bolts on the cover with an uniform and balanced force to avoid affecting its tightness.

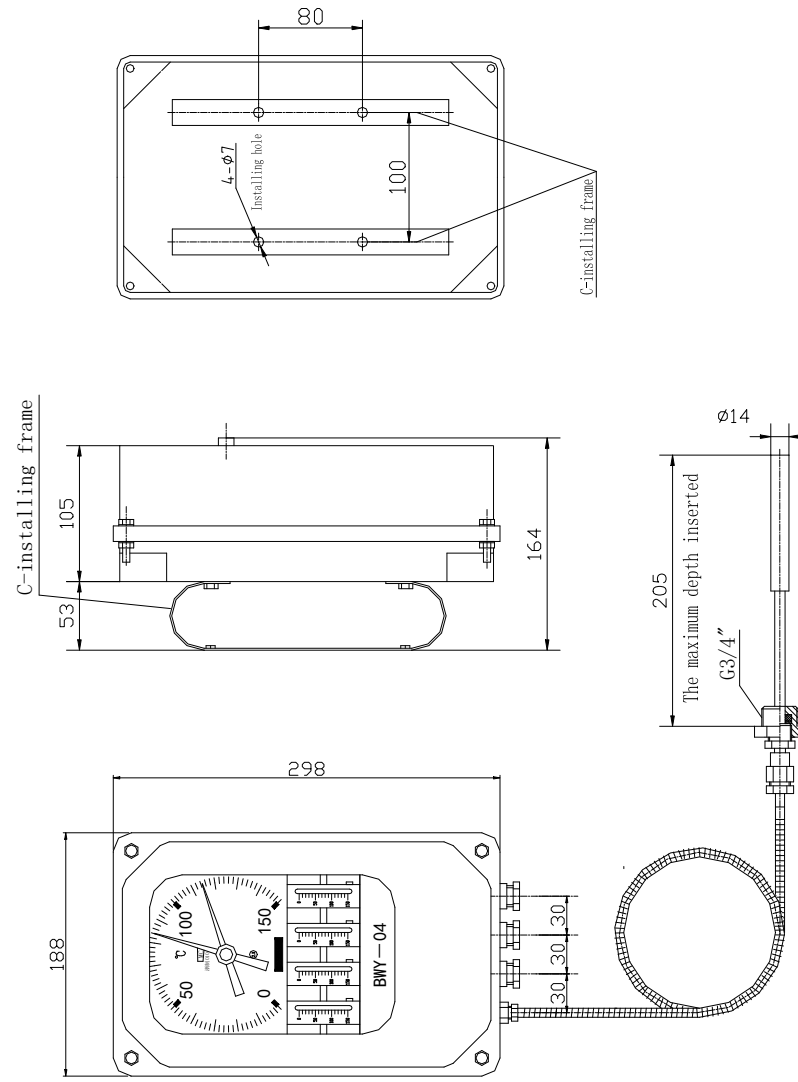
4. While installing the unit, the surplus capillary tube should be wound into a circle of diameter of over 150mm and properly fixed at every interval of 300mm (for example tied by some thin copper wire etc).

5. The test on the Digital Display Thermometer can be conducted by connecting it either with a standard direct current signal source or with a BWR—04 with its bulb inserted into a constant temperature trough.

6. During field installation attention must be paid to insert the entire bulb into the transformer's sleeve with oil inside it. In other words, the depth of inserted bulb should be at least no less than 150mm.

7. Water is forbidden to let in the transformer's sleeve with oil inside. When the ambient temperature drops below 0℃ and the transformer happens not to work. The water in the sleeve probably freezes in an expansion to squeeze the bulb, which results in wrong indication of the thermoregulator. So the attention must be paid to seal the bulb when mounting it to prevent rainwater in.

8. If use the situation do not need the digital display thermometer should install the DC24V power source, used to the transmitter inside of the winding thermoregulator changes to send the operating power supply.



Addendum The outline drawing and mounting dimension of the export of BWR-04